

Answering Lord of the Rings Trivia with Large Language Models

Three Approaches: Zero-Shot, Fine-Tuned Prediction, and Synthetic-Data Training

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1 Introduction

Large language models have demonstrated impressive capabilities on a wide range of knowledge-intensive tasks. However, their performance on highly specific fictional lore—such as the detailed world-building of Tolkien’s Middle-earth—is less well studied. This project uses a curated set of LOTR trivia questions as a benchmark to compare three distinct LLM adaptation strategies:

1. **Zero-shot prompting** (Approach A): a state-of-the-art instruction-tuned LLM is queried with no task-specific training, relying solely on knowledge acquired during pre-training.
2. **Retrieval-augmented generation** (Approach B): a general-purpose LLM is paired with a retrieval module that indexes LOTR source text into a vector store; at inference time, the top-k passages most semantically similar to the query are retrieved and injected into the prompt as context, allowing the model to ground its answer in retrieved evidence without any parameter updates.
3. **Fine-tuned next-word prediction** (Approach C): a 7B parameter base language model with a publicly available LoRA adapter trained on LOTR text is evaluated in a question-answering prompt format.
4. **Synthetic-data fine-tuning** (Approach D): an LLM is first used to generate a synthetic dataset of LOTR question-answer pairs, which is then used to fine-tune a second model specifically for this task.

All experiments run on hardware accelerated by NVIDIA CUDA GPUs, allowing efficient inference with multi-billion-parameter models.

2 Dataset

2.1 PubMedQA

Initially, the PubMedQA dataset was considered as the basis for this thesis’s experiments. However, it proved unsuitable for our purposes: the questions are highly technical and require substantial domain expertise to evaluate meaningfully, an area in which we lacked the necessary background. Additionally, each question in PubMedQA is already paired with its supporting abstract as context, which would have rendered the retrieval step in a RAG pipeline largely redundant. For these reasons, we opted instead for a dataset based on the Lord of the Rings novels, described below.

2.2 Source and composition

The benchmark dataset contains a total of 169 trivia questions about *The Lord of the Rings*.

2.3 Answer format

Each question has a single short-form reference answer (typically one to five words, e.g. “*Frodo Baggins.*” or “*Strider.*”). The answers are not multiple-choice; the model must generate them freely.

3 Evaluation Methodology

3.1 Answer extraction

After generation, the raw model output is post-processed as follows:

1. If the text contains the substring "Answer:", everything before (and including) the first occurrence is discarded.
2. The first line of the remaining text is taken.
3. The first sentence (split on ., !, ?) of that line is used as the predicted answer.

3.2 Scoring

A prediction is counted as correct (*hit*) when a case-insensitive, whitespace-normalised substring match holds in either direction:

$$\text{hit} = 1 \iff \hat{a} \subseteq a \text{ or } a \subseteq \hat{a},$$

where a is the normalised ground-truth answer and $\hat{a}$ is the normalised prediction. The final metric is $\text{accuracy} = \text{hits}/N$.

4 Approach A: Zero-Shot LLMs

4.1 Models evaluated

Two instruction-tuned models were evaluated in a zero-shot setting, receiving only the question and a fixed system prompt, with no fine-tuning or retrieved context provided. **Qwen3.5-2B** and **Gemma-4-E2B-it** were chosen as comparably sized (2B-parameter) general-purpose models, allowing a comparison of pre-trained LOTR knowledge across

model families at a similar scale. Both models were queried with an identical system prompt instructing them to answer concisely, without explanation:

"You are a Lord of the Rings trivia answerer. Reply with ONLY the answer to the question. No extra words, no full sentences, no explanation."

This constraint was used to simplify answer extraction and enable direct comparison against the reference answers in the evaluation set, without requiring additional parsing or normalization of free-form responses.

4.2 Results

Model	Accuracy (%)
Qwen3.5-2B	12.3
Gemma-4-E2B-it	44.3

Table 1: Zero-shot accuracy on the LOTR QA evaluation set.

Gemma-4-E2B-it clearly outperformed Qwen3.5-2B (44.3% vs. 12.3%), though the exact cause of this gap is unclear given the comparable model sizes. Nonetheless, both models demonstrate some baseline knowledge of LOTR content, indicating that the task is not fundamentally out of hand for models of this scale, while also leaving substantial room for improvement, which the retrieval- and fine-tuning-based approaches (B–D) aim to address.

5 Approach B: Retrieval Augmented Generation (RAG)

5.1 Chunking strategy

The source corpus for retrieval consisted of the full text of the Lord of the Rings novels. The raw text was first normalized by collapsing all whitespace (newlines, tabs, repeated spaces) into single spaces, removing formatting artifacts from the original source files. The cleaned text was then split into overlapping chunks using a fixed-size sliding window over whitespace-tokenized words: each chunk consisted of 25 words, with a 5-word overlap between consecutive chunks (a step size of 20 words per chunk). This overlap was intended to reduce the risk of relevant information being split across a chunk boundary and lost during retrieval. The relatively short chunk size of 25 words was chosen to keep retrieved passages narrow and topically focused, which is particularly suited to fact-style trivia questions where the relevant answer is often contained in a single sentence or clause rather than spread across a long passage.

5.2 Embedding model

Chunks were embedded using **BAAI/bge-base-en-v1.5**, a general-purpose sentence embedding model from the BGE family, run on GPU and held in memory for the duration of evaluation (no persistent vector database was used). Document chunks were embedded directly, with embeddings L2-normalized so that retrieval similarity could be computed as a simple dot product, equivalent to cosine similarity. Queries, in contrast, were embedded with BGE’s recommended query instruction prefix (*"Represent this sentence for searching relevant passages: "*) prepended to the question text, following the model’s documented usage pattern for asymmetric retrieval, where queries and documents are encoded with different prompting conventions to better align the two embedding spaces. At retrieval time, the top- k chunks (with $k = 3$ used in evaluation) with the highest cosine similarity to the query embedding were selected as context.

5.3 Answering pipeline

For answer generation, **Gemma-4-E2B-it** was used as the generator model, loaded via Hugging Face’s `AutoModelForCausalLM` and `AutoProcessor` interfaces. For each question, the top- k retrieved chunks were concatenated and inserted into a fixed prompt template alongside the question, under a system prompt instructing the model to answer using only the provided context and to reply with the answer alone, without additional explanation or full sentences:

"You are a Lord of the Rings trivia answerer. Use the provided context to answer. Reply with ONLY the answer to the question. No extra words, no full sentences, no explanation."

The prompt was assembled using the model’s chat template and passed to the model for generation under greedy decoding (`do_sample = False`), with generation capped at 40 new tokens to keep responses short and consistent with the single-answer response format used throughout the evaluation.

5.4 Results

Model	Zero-shot (%)	RAG (%)	Δ
Qwen3.5-2B	12.3	24.5	+12.2
Gemma-4-E2B-it	44.3	29.2	−15.1

Table 2: Accuracy on the LOTR QA evaluation set with and without retrieval-augmented generation.

Adding retrieval had opposite effects on the two models: Qwen3.5-2B roughly doubled its accuracy (12.3% $\rightarrow$ 24.5%), while Gemma-4-E2B-it’s accuracy dropped substantially (44.3% $\rightarrow$ 29.2%). This divergence highlights a key property of RAG pipelines: overall answer accuracy is upper-bounded by retrieval accuracy, since a generator can only answer correctly from context that actually contains the relevant fact. For Qwen3.5-2B, whose zero-shot pre-trained knowledge of LOTR content was weak, even imperfect retrieved context provided a net benefit. For Gemma-4-E2B-it, which already encoded substantial LOTR knowledge internally, irrelevant or noisy retrieved chunks appear to have distracted the model or displaced correct internal knowledge, resulting in a net loss relative to the zero-shot baseline.

This suggests that the current retrieval step, rather than the generator, is the primary bottleneck limiting RAG performance in this setup. Several directions could improve retrieval accuracy and, in turn, downstream answer accuracy:

- **Query rewriting:** short or ambiguously phrased trivia questions may embed poorly relative to the longer, more narrative phrasing of the source chunks; rewriting the query into a more passage-like statement before embedding could improve alignment between query and document embedding spaces.
- **Hybrid search:** combining dense embedding-based retrieval with sparse lexical retrieval (e.g. BM25) could help surface chunks containing exact names, places, or terms that dense embeddings alone may under-weight, particularly for proper nouns specific to the LOTR universe.
- **Reranking:** retrieving a larger candidate set of chunks and applying a cross-encoder reranker to reorder them by relevance before selecting the final top- k context could reduce the inclusion of superficially similar but non-relevant chunks, directly addressing the noise that likely degraded Gemma-4-E2B-it’s performance.

6 Approach C: Fine-Tuned Next-Word Prediction

6.1 Model and adapter

The base model used is **DeepSeek-LLM-7B-Base** ([deepseek-ai/deepseek-llm-7b-base](#)) [1], a 7-billion parameter causal language model. A LoRA adapter ([cwestbrook/lotrdata](#) [2]) trained on LOTR prose text is loaded on top of the frozen base using the PEFT library [3]. The adapter was designed for next-word prediction on Tolkien text rather than explicit question answering.

6.2 Results

The fine-tuned model answered correctly 21 out of 106 questions, yielding an overall accuracy of **19.8%** (85 incorrect predictions). Table 3 shows representative correct and incorrect predictions.

Table 3: Selected predictions from Approach B.

Question	Gold Answer	Prediction	Hit
Which Elf joins the Fellowship as archer?	Legolas	Legolas	✓
Which Dwarf represents his people in the Fellowship?	Gimli	Gimli	✓
Who is the Ring-bearer for most of the journey?	Frodo Baggins	Frodo	✓
Aragorn travels under what ranger alias when first met?	Strider	Shadowfax	×
Frodo’s most steadfast companion is?	Samwise Gamgee	Bilbo Baggins	×
What is Pippin’s full name?	Peregrin Took	Shadowfax	×
What is Gandalf called after his return?	Gandalf the White	The horse is called Shadowfax	×
Bilbo gifts Frodo a shirt of what metal?	Mithril	The River	×
Saruman’s obsidian tower is called?	Orthanc	Bag End	×

6.3 Failure mode analysis

Three recurring failure modes were identified:

1. **Next-word continuation drift.** Because the adapter was trained on narrative prose rather than QA pairs, the model frequently continues generating additional “Question:/Answer:” turns instead of stopping after the first answer. The answer extraction heuristic mitigates this in some cases but not all.
2. **Hallucinated *Shadowfax* responses.** The word “Shadowfax” (Gandalf’s horse) appears as the predicted answer for at least five unrelated questions spanning multiple categories. This suggests the model’s fine-tuning corpus may have over-represented passages featuring Shadowfax, creating a spurious high-prior token.
3. **Unicode and diacritic mangling.** Several predictions contain garbled diacritics (e.g. Barad-dûr, Gloín), indicating that the tokeniser or decoding pipeline does not faithfully reconstruct special characters that appear in the original Tolkien text.

7 Approach D: Synthetic QA Generation and Fine-Tuning

7.1 Q&A generation

To construct a training set of question–answer pairs grounded in the source novels, the full LOTR text was first split into smaller fragments, analogous to the chunking step used for retrieval in Approach B. Each fragment was then passed to an AI application built as a Claude Artifact, which was prompted to generate trivia-style question–answer pairs based strictly on the content of the fragment. To reduce the number of API calls and conserve tokens, five text fragments were batched into a single prompt at a time, with the model instructed to generate one question–answer pair per fragment (five pairs per prompt) rather than issuing a separate request per fragment. Pairs were returned in a fixed JSON format to allow automated parsing and aggregation into the final dataset, for example:

```
{
  "question": "What is the elvish name for the location also called
the Pillars of the Kings or Gates of Gondor?",
  "answer": "Argonath"
},
{
  "question": "What is the common name for the healing herb known
in elvish as aséa aranion?",
  "answer": "Kingsfoil"
},
{
  "question": "Who was Primula Baggins's maiden family name before
she married into the Baggins family?",
  "answer": "Brandybuck"
},
{
  "question": "What is the elvish name for the Barrow-downs, also
called Tyrn Gorthad?",
  "answer": "Tyrn Gorthad"
},
{
  "question": "What legendary hero is known by the epithet
'One-hand'?",
  "answer": "Beren"
```

```

},
{
  "question": "What nickname was Otto Boffin known by?",
  "answer": "the Fat"
},

```

This batched generation process was repeated across the full set of text fragments derived from the novels, yielding a total of **2,389** question–answer pairs, which formed the synthetic dataset used for fine-tuning in this approach.

7.2 Fine-tuning

Qwen/Qwen2.5-3B-Instruct was fine tuned using Low-Rank Adaptation (LoRA) with rank 16, applying adapters to attention and feed-forward layers (29.9M trainable parameters, 0.96% of total). Training used the SFT objective on the Q/A pairs (80/20 train/test split), with an effective batch size of 16 in `bfloat16` precision. A cosine learning rate schedule ($\text{lr} = 0.0002$, 50 warmup steps) was used for 15 training epochs.

8 Comparative Results

Approach	Model / Configuration	Accuracy (%)
A: Zero-Shot	Qwen3.5-2B	12.3
	Gemma-4-E2B-it	44.3
B: RAG	Qwen3.5-2B	24.5
	Gemma-4-E2B-it	29.2
C: Fine-Tuned (Next-Word)	DeepSeek-LLM-7B-Base + LoRA	19.8
D: Fine-Tuned (Synthetic QA)	Qwen2.5-3B-Instruct + LoRA (15 epochs)	32.0

Table 4: Overall accuracy across all evaluated approaches on the LOTR trivia dataset.

References

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- [4] T. Dettmers et al., “LLM.int8(): 8-bit Matrix Multiplication for Transformers at Scale,” *NeurIPS*, 2022. <https://github.com/TimDettmers/bitsandbytes>
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